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EDUCATION

Stony Brook School, NY, USA | Grades 9-12 August 2023 - present
High School Diploma, Expected May 2027

- **Math related courses**: Pre-Calculus Honors (A+), AP Calculus BC (A), College Multivariable Calc (A+), Advanced Linear Algebra(projected grade: A+)
- **APs**: Calculus BC (5), English Language & Composition (4), European History (4), Macroeconomics (5), Microeconomics (5), AP Physics C: Mechanics (4), Statistics (4)
- **SAT**: 1520 (Reading and Writing: 730, Math: 790)

AWARDS AND HONORS

Grade 11

- USAMO – Qualifier (for scoring 13 in AIME)
- AMC12A – perfect score (150/150)
- USACO – promoted to Platinum Division in February Contest
- AMC12A – Honor Roll of Distinction (for scoring in top 1%)
- HiMCM 2025 – Honorable Mention
- AIME – promoted to competition by MAA through AMC12
- MathWorks Math Modeling Challenge (M3) – Successful in Triage
- Multi-Variable Calculus Student of the Year
- High Honor Roll (for academic performance, fall and spring semesters)

Grade 10

- AIME – promoted to competition by MAA through AMC12
- Public Benefit International Challenge for Youth (PBIC) – team First Prize
- AMC12A – Certificate of Distinction (for scoring in top 5%)
- China Conrad Challenge – Finalist
- High Honor Roll (for academic performance, fall semester)
- Head of School Honor Roll (for academic performance, spring semester)

Grade 9

- AMC10A - Honor Roll of Distinction (for scoring in top 1%)
- AIME – promoted to competition by MAA through AMC10
- AMC10B – Certificate of Distinction (for scoring in top 5%)
- High Honor Roll (for academic performance, fall and spring semesters)

SUMMER PROGRAMS

Computer Science and Informatics Summer Research Experience Program (CSIRE) at Stony Brook University| *Summer 2026*

- Selected for college-level computer science research program.
- Conducted research under Professor Shuai Mu on Mako: Speculative Distributed Transactions with Geo-Replication.
- Modeled/compared distributed transaction systems, geo-replication, traditional two-phase commit and speculative execution.

- Presented research findings in final paper, poster, and symposium presentation.

Pioneer Academics Summer Term | *Summer 2025*
- Completed intensive concentration in Graph Decomposition & Word Avoidance (Mathematics).
- Conducted independent research on graph theory, focusing on bi-clique decompositions and edge contraction.
- Investigated how edge contraction affects the minimum number of complete bipartite graphs required to decompose a graph.
- Wrote research paper on findings.

AwesomeMath | *Summer 2024*
- Completed Computational Geometry – Level 2, an advanced problem-solving course in geometry.
- Studied advanced geometric techniques and approaches to competition-level problems.
- Collaborated with peers to develop and compare multiple problem-solving strategies.

IVY LEAGUE SUMMER SCHOOL | *Summer 2023*
- Completed *English Literature and Writing* and *Public Speaking* courses.

RESEARCH

Differential Privacy | *Grades 11 and 12*
- Compared local and central differential privacy in online advertising systems.
- Simulated 10,000 users and 88,000+ ad impressions across eight campaigns.
- Implemented and evaluated three differential privacy mechanisms in Python.
- Tested privacy/utility tradeoffs across privacy budgets, dataset sizes, and click rates.
- Evaluated accuracy, campaign ranking, and attribute-inference privacy risks.
- Wrote 8000+-word research paper (published in National High School Journal of Science [NHSJS], link: https://nhsjs.com/2026/under-what-conditions-is-local-differential-privacy-preferable-to-central-differential-privacy-a-synthetic-study-of-online-advertising-campaign-measurement/).
- Mentor: Prof. Andrew Blumberg (Biology, Computer Science, and Mathematics, Columbia University).

Environmental Footprint Modeling and Sustainability Ranking of Mega Sporting Events | *Grade 11*
- Led 2-person team in High School Mathematical Contest in Modeling (HiMCM). Conducted quantitative modeling project on evaluating the environmental impacts of mega sporting events, focusing on energy consumption, greenhouse gas emissions, water usage, and waste generation.
- Developed hierarchical sub-models for each indicator and applied an entropy/TOPSIS framework to rank host cities by sustainability performance; validated the model using Super Bowl LIX data and extended it to multi-stage international tournaments (BWF World Tour);
- Wrote 25-page research report.

High-Energy Computing Environmental Footprint | *Grade 10*
- Led two-person team in this project. Conducted quantitative modeling project to assess the environmental impact of High-Performance Computing (HPC), analyzing global data centers' energy consumption, carbon emissions, and water usage.

- Developed predictive models (Auto Regressive Integrated Moving Average [ARIMA], Long Short-Term Memory [LSTM]) to forecast future trends and proposed sustainable solutions.
- Wrote 20+-page research report.

Medicine AI - Intelligent Medication Assistant for Elderly Care | *Grade 10*
- For Conrad Challenge, developed AI-powered medication assistant for elderly and chronic-disease patients.
- Integrated OCR and TextCNN to process medical records and generate personalized health suggestions.
- Designed medication reminders, pill-image alerts, and daily health reports for users and caregivers.
- Tested system using 500+ medical data sets and developed user-friendly interface with PyQt5.

INTERNSHIP

Rongdingyue (Zhejiang) Aviation Technology Co., Ltd. | Zhejiang, China
UAV Technology Data Analysis & International Business Intern | *Summer 2026*
- Spent two weeks assisting with Unmanned Aerial Vehicle (UAV) technology data analysis and international business activities in an aviation-technology company.
- Updated and Modified English version of company's Product Manual.
- Operated UAVs and managed flight-test data, evaluating maximum payload capacity and analyzing real-world performance data for new drone models.
- Served as UAV flight test operator, conducting pre-delivery test flights.

EXTRACURRICULAR ACTIVITIES

Chinese Leader | *Grade 12*
- Represent/support 100+ Chinese students
- Inform parents about important school information.
- Update/revise school's Chinese language handbook.

E-sports Club | Vice President | *Grade 11 - present*
- Organize 5+ school-wide e-sports (= gaming) tournaments and events: Clash Royale, Counter-Strike, etc.
- Lead gaming discussions and workshops focused on strategy, teamwork, and skill development.
- Help oversee monthly meetings of 30+-member club.

Asian Student Union (ASU) | President (Grade 12), Executive Board Member (Grade 11), Member (Grades 9 and 10)
- Lead 10-member Executive Board and 30+ ASU members in organizing cultural programs and other initiatives
- Organized 6 cultural events, generating $2,000+ in funds for ASU programs.

Chinese Teaching Program | Teaching Assistant | *Grade 9 - present*
- Assistant teach Chinese language and culture, twice per month, to approximately 20 elementary school students as employee of Center for Chinese Learning at Stony Brook

Chess Club | Member | *Grade 9 - present*
- Competitive chess player with experience in regional and international tournaments.
- Support newer players through strategy analysis, game review, and tournament preparation.

- Helped school team win Suffolk County Team Championship in Grades 9 and 10.

Sports-related Injuries Program| *Grade 9 - 11*
- Co-led student-led initiative to reduce sports-related injuries among youth athletes in Tanzania.
- Raised approximately $5,000 to purchase and donate knee pads, ankle braces, wrist guards, and other protective equipment.
- Partnered with two soccer clubs in Arusha, Tanzania to distribute protective equipment to dozens of young athletes.
- Collaborated with local medical professionals to create injury-prevention resources and educational materials.

Admissions Council | Tour Guide | Student Ambassador | *Grade 10 - present*
- Worked as one of 70 student ambassadors in my school's Admissions Council.
- Led 10+ tours for new students and their families.

Kitchen Crew Captain | Workjob Captain | *Grade 11 - present*
- Served as kitchen dishwasher.
- Lead 20 students in weekly tradition of serving the school community through kitchen work.

OTHER

- **Technical Skills:** C++, Python, Mathematical Modeling, Data Analysis
- **Languages:** Native Mandarin, Fluent English, Intermediate French
- **Certificates:** Piano Level 7; Taekwondo Blue-Red Belt
- **Hobbies:** Badminton, Basketball, Cycling, Photography